QUESTION 1 [2 + 2 + 4 + 2 = 10 MARKS]

Kei Nishikori is an asset manager based in Tokyo, Japan and manages the Grand Slam Fund, a portfolio of about 50 Japanese stocks, whose current value is 12 billion yen. Over the past twelve months, Kei has kept the beta of the portfolio rather low at 0.3 as he attached a high probability to a large drop in the Japanese stock market.

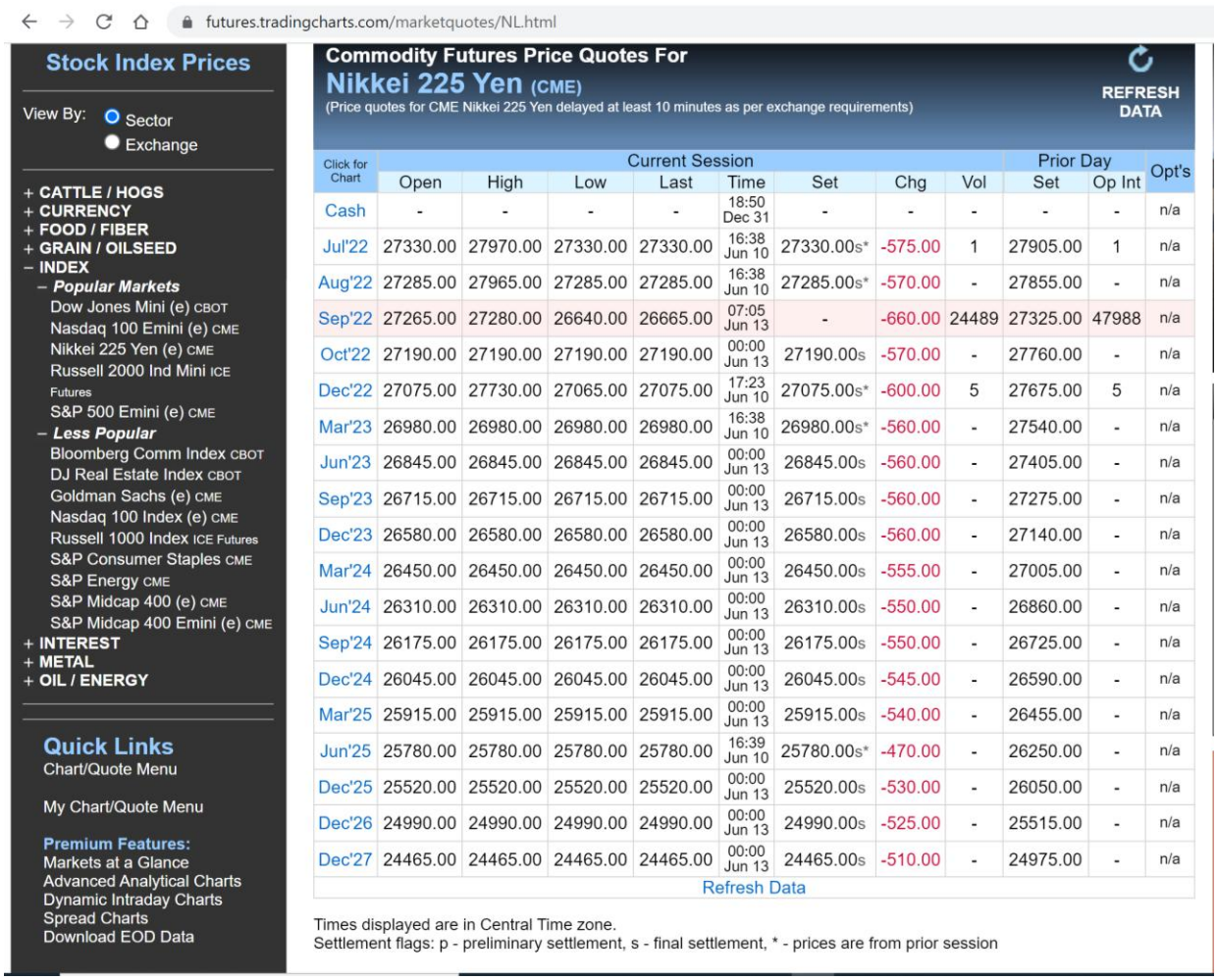
Below are some market information and performance statistics over the past year:

	Performance Measures			
	Return	Risk (std dev)	Beta	Alpha
Grand Slam Fund	-2.80%	18%	0.3	0.42%
NIKKEI 225 Spot	- 11.30%	21.2%	1	0%
NIKKEI 225 Futures	- 11.10%	20.9%	1	0%
Japan Risk Free rate (current as of 13/06/2022): 0.25%				
NIKKEI 225 Spot Level as of 13/06/22): 26,987				
Note: - The return, risk (as measured by the standard deviation of returns), beta and alpha refer to the 13/06/2021- 13/06/2022 period - The Risk free rate is per annum and continuously compounded				

	Correlation Matrix of Returns – Annualized		
	NIKKEI 225 Spot	NIKKEI 225 Futures	GRAND SLAM Fund
NIKKEI 225 Spot	1		
NIKKEI 225 Futures	0.98	1	
GRAND SLAM Fund	0.65	0.62	1

As of today, June 13, 2022, Kei and his team believe that the downward correction is over and the Japanese market is ready for a run-up over the next 6 months. In facts, he expects the Nikkei 225 to generate a 7% return (including dividend yield) over the next six months, while he believes his portfolio will deliver the same alpha (i.e., 0.42%/2=0.21% over six months). The investment committee (IC) of the Grand Slam fund agrees with Kei: they, thus, decide to increase the exposure of the fund to the overall equity market. They decide that an overall beta of 1.5 is their goal and that futures contracts on the Nikkei 225 are the appropriate tool to reach such a goal.

Kei looks up some quotes for Nikkei futures. Below is what he sees on his screen:



futures.tradingcharts.com/marketquotes/NL.html

Stock Index Prices

View By: Sector / Exchange

+ CATTLE / HOGS
+ CURRENCY
+ FOOD / FIBER
+ GRAIN / OILSEED
– INDEX
  – *Popular Markets*
    Dow Jones Mini (e) CBOT
    Nasdaq 100 Emini (e) CME
    Nikkei 225 Yen (e) CME
    Russell 2000 Ind Mini ICE Futures
    S&P 500 Emini (e) CME
  – *Less Popular*
    Bloomberg Comm Index CBOT
    DJ Real Estate Index CBOT
    Goldman Sachs (e) CME
    Nasdaq 100 Index (e) CME
    Russell 1000 Index ICE Futures
    S&P Consumer Staples CME
    S&P Energy CME
    S&P Midcap 400 (e) CME
    S&P Midcap 400 Emini (e) CME
+ INTEREST
+ METAL
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Commodity Futures Price Quotes For
Nikkei 225 Yen (CME)
(Price quotes for CME Nikkei 225 Yen delayed at least 10 minutes as per exchange requirements)
REFRESH DATA

Click for Chart	Current Session								Prior Day		Opt's
	Open	High	Low	Last	Time	Set	Chg	Vol	Set	Op Int	
Cash	-	-	-	-	18:50 Dec 31	-	-	-	-	-	n/a
Jul'22	27330.00	27970.00	27330.00	27330.00	16:38 Jun 10	27330.00s*	-575.00	1	27905.00	1	n/a
Aug'22	27285.00	27965.00	27285.00	27285.00	16:38 Jun 10	27285.00s*	-570.00	-	27855.00	-	n/a
Sep'22	27265.00	27280.00	26640.00	26665.00	07:05 Jun 13	-	-660.00	24489	27325.00	47988	n/a
Oct'22	27190.00	27190.00	27190.00	27190.00	00:00 Jun 13	27190.00s	-570.00	-	27760.00	-	n/a
Dec'22	27075.00	27730.00	27065.00	27075.00	17:23 Jun 10	27075.00s*	-600.00	5	27675.00	5	n/a
Mar'23	26980.00	26980.00	26980.00	26980.00	16:38 Jun 10	26980.00s*	-560.00	-	27540.00	-	n/a
Jun'23	26845.00	26845.00	26845.00	26845.00	00:00 Jun 13	26845.00s	-560.00	-	27405.00	-	n/a
Sep'23	26715.00	26715.00	26715.00	26715.00	00:00 Jun 13	26715.00s	-560.00	-	27275.00	-	n/a
Dec'23	26580.00	26580.00	26580.00	26580.00	00:00 Jun 13	26580.00s	-560.00	-	27140.00	-	n/a
Mar'24	26450.00	26450.00	26450.00	26450.00	00:00 Jun 13	26450.00s	-555.00	-	27005.00	-	n/a
Jun'24	26310.00	26310.00	26310.00	26310.00	00:00 Jun 13	26310.00s	-550.00	-	26860.00	-	n/a
Sep'24	26175.00	26175.00	26175.00	26175.00	00:00 Jun 13	26175.00s	-550.00	-	26725.00	-	n/a
Dec'24	26045.00	26045.00	26045.00	26045.00	00:00 Jun 13	26045.00s	-545.00	-	26590.00	-	n/a
Mar'25	25915.00	25915.00	25915.00	25915.00	00:00 Jun 13	25915.00s	-540.00	-	26455.00	-	n/a
Jun'25	25780.00	25780.00	25780.00	25780.00	16:39 Jun 10	25780.00s*	-470.00	-	26250.00	-	n/a
Dec'25	25520.00	25520.00	25520.00	25520.00	00:00 Jun 13	25520.00s	-530.00	-	26050.00	-	n/a
Dec'26	24990.00	24990.00	24990.00	24990.00	00:00 Jun 13	24990.00s	-525.00	-	25515.00	-	n/a
Dec'27	24465.00	24465.00	24465.00	24465.00	00:00 Jun 13	24465.00s	-510.00	-	24975.00	-	n/a

Refresh Data

Times displayed are in Central Time zone.
Settlement flags: p - preliminary settlement, s - final settlement, * - prices are from prior session

Note 1: Assume that Kei can trade at the last quoted prices and the index multiplier for the above contracts is ¥650.
Note 2: The expiration date is the second Friday of the contract month.

Based on the above information:

a) What futures position should the Grand Slam fund take to achieve the goal stated above? Explain your answer.

Note: you are not required to work out the number of contracts in this part of the question, only the direction of the trade.

2 MARKS

Portfolio is naturally LONG and has positive beta w.r.t. Nikkei 225: → To increase the beta, futures position must be LONG as well

b) Based on the available futures, Kei decided to use the March 2023 contract. How many contracts should Kei trade to reach the goal stated above? **Show all your detailed workings and round the final answer to the nearest integer.**

2 MARKS

Yen (¥) Value of one Futures March Contract is =¥650*26,980 = ¥17,537,000

Portfolio Value as of 13/06/2022 = ¥ 12,000,000,000

Kei should go long (1.5 – 0.3)*12,000,000,000 /17,537,000
=821

i.e., 821 Nikkei 225 futures contracts

c) With the above strategy in place, what risks (if any) will the Grand Slam fund be exposed to until expiration of the futures position? With the futures position in place, is the fund more or less exposed to those risks than before taking the position? Explain your answers.

As the beta of the portfolio is different from zero, the portfolio is still exposed to equity market risk.

1 MARK

Furthermore, the strategy still leaves the portfolio exposed to basis risk because of

1. asset mismatch: the asset being hedged (i.e., Grand Slam portfolio), is not the same as the asset underlying the futures contract (which is the Nikkei 225 index)
2. Maturity mismatch

2 MARKS

Actually, after the fund takes on the futures position, the overall (stocks + futures) position is more exposed to market risk than before taking the futures

1 MARK

d) The IC asks Kei what return the hedged portfolio is expected to earn during the 6-month investment period. Given the strategy you suggest above and given all the information

provided, what percentage return should Kei expect to earn on his overall position during the next six months? **Show all your detailed workings.**

Note: you can assume that the riskfree rate over the upcoming six-months is half of the annualized rate reported above (i.e, 0.25%/2 = 0.125%)

2 MARKS

The futures strategy aims to make the beta of the hedged portfolio equal to 1.5. The strategy assumes the CAPM model. Given the estimates in the table above and expectations about the market return, the expected return on the stock = futures portfolio equals the risk-free rate + the alpha of the portfolio + beta*(expected market risk premium). I.e., 0.25%/2 + 0.42%/2 + 1.50*(7% - 0.25%/2)= 10.65% over the 6-month investment period

QUESTION 2 [1+2+2+2 = 7 MARKS]

A Reuters article covered in class reported that an oil-price shock linked to Middle East geopolitical tensions increased the cost for Indian importers to hedge U.S. dollar payments.

Assume you are an Indian importer with a large U.S. dollar payable due in six months. In the recent past, you've been hedging U.S. dollar payables using a hedge ratio of 50%.

a) What exchange-rate risk do you face on the USD payable due in six months? What forward position would hedge this risk?

b) The article notes that USD/INR forward premiums increased. Explain, without doing any numerical calculation, why such increase makes a forward hedge more costly for you.

c) The article further points out that the implied volatility of currency options rose as well. Explain why this increase translates into a costlier hedge for you.

d) Discuss how your expectation about the duration of the geopolitical shock may affect your decision to increase or not your hedge ratio from current levels.

NOTE: all answers and discussion for this question don't need calculations. Conceptual answers are sufficient to get full marks.

SOLUTION

a) An importer needs to buy U.S. dollars in the future. They are exposed to USD appreciation (i.e, INR depreciation). The appropriate forward hedge is to buy USD forward, or equivalently go long a USD/INR forward contract.

b) Higher USD/INR forward premium means the forward rate is further above the spot rate. Since the importer must buy dollars, a higher forward rate means it locks in a larger rupee payment for each dollar of future liability. The hedge still reduces exchange-rate uncertainty, but the locked-in rate is less favourable.

c) Higher implied volatility means larger possible exchange-rate movements. A U.S. dollar call option becomes more valuable because it protects the importer if the dollar rises sharply, while preserving limited downside for the option holder.

d) The hedge ratio is the proportion of the USD payable that the importer hedges. If the conflict is expected to be prolonged, the importer may increase the hedge ratio because the risk of rupee depreciation is likely to persist until the payment date. The higher hedge ratio reduces the chance that the firm must buy USD at an even worse spot rate later. However, because forward premiums and option implied volatility have risen, hedging is more expensive. If the conflict is expected to be short-lived, the importer may prefer a lower hedge ratio because hedging costs may fall and the rupee may recover before the payable comes due.

QUESTION 3 [2 + 3 + 2 = 7 MARKS]

It is Monday, April 20, 2026. Elaine Benes, senior portfolio manager at YadaYada Pension Management, manages a U.S. growth-equity portfolio for a large superannuation client. The client currently holds a USD 80 million physical equity portfolio designed to mirror the Nova Growth 100 Index (NG100), a price index composed of U.S. growth stocks. The stocks in the NG100 Index pay no dividend over the next six months and the index has no other income/carrying costs apart from financing.

Elaine receives a dealer quote for a six-month cash-settled forward contract on the NG100. Each forward contract is written on 100 index units. The forward perfectly replicates the underlying index and settles in cash at maturity. The long side, thus, receives 100*(S_T – F_dealer), where S_T is the NG100 index level at maturity and F_dealer is the forward delivery price.

On her computer screen Elaine sees the following market information:

Item	**Value**
Current NG100 index level	2,000
Dealer forward quote	2,025
Time to maturity of Forward	0.5 years
Risk-free interest rate	3.8% p.a., continuously compounded
Dividend yield on NG100 Index	0%

Assume there are no taxes, transaction costs, bid-ask spreads, margin requirements, or short-selling constraints. It follows that opening a forward position at t=0 does not generate any cash inflow or require any cash outflow.

YadaYada's investment committee does not want Elaine to make a speculative bet on the future direction of the index. However, it allows her to temporarily replace the physical portfolio with a derivative position if the strategy restores the client's NG100 exposure at maturity and generates a riskless profit.

a) [2 marks] Using the information above, show why the riskless profit opportunity exists.

b) [3 marks] Describe the arbitrage strategy Elaine should implement at t = 0. Specifically, show what trades she needs to make, the cash flow associated with each trade and the total net cash flow at t=0.

Hint: As part of the strategy, you will need to compute how many forward contracts Elaine should use to replace the NG100 exposure of the USD 80 million portfolio.

c) [2 marks] Show the trades and cash flows at maturity of the forward and the arbitrage profit.

SOLUTION

a)

For an underlying asset with no income, no dividends, and no storage cost, the no-arbitrage forward price is:

$F_0 = S_0 e^{rT}$

$F_0 = 2{,}000\ e^{0.038 * 0.5} = 2{,}000\ e^{0.019} = 2{,}038.36$

The dealer quote is $F_{dealer} = 2{,}025$, which is below 2,038.36. Therefore, the forward contract is underpriced by:

2,038.36 - 2,025 = 13.36 index points

This mispricing is why the arbitrage opportunity exists.

b)

Because the forward is underpriced, Elaine should: sell the physical index portfolio, invest the sale proceeds at the risk-free rate, and enter into a long position in the underpriced forward contracts.

The USD 80 million portfolio represents the following number of index units:

80,000,000 / 2,000 = 40,000 index units

Each forward contract is written on 100 index units, so Elaine should use:

40,000 / 100 = 400 contracts

➔ Elaine should open a long position in 400 NG100 forward contracts.

At t = 0:

Action	**Cash flow at t = 0**	**Comment**
Sell the physical NG100 portfolio	+USD 80,000,000	The client gives up the physical portfolio temporarily.
Enter 400 long forward contracts	USD 0	A forward contract has zero value at inception.
Invest sale proceeds at the risk-free rate	-USD 80,000,000	The proceeds are invested for six months.
Net t = 0 cash flow	USD 0	No initial wealth is required beyond the existing portfolio.

NOTE: If this was an actual exam question, the Comment column in the table above wouldn't be required. It is added here to clarify things further.

c)

At maturity (t=T)

The risk-free investment is worth:

$80{,}000{,}000\ e^{0.038 * 0.5} = 81{,}534{,}531.89$

The payoff from 400 long forward contracts is:
400 * 100(S_T - 2,025) = 40,000*(S_T - 2,025)
= 40,000*S_T - 81,000,000
To restore the client's physical NG100 exposure, Elaine repurchases 40,000 index units at maturity:
Cost of repurchase = 40,000*S_T

The final cash left after restoring the portfolio is therefore:
81,534,531.89 + [40,000S_T - 81,000,000] - 40,000S_T
= 534,531.89

The arbitrage profit is USD 534,531.89. It does not depend on S_T, so the strategy is riskless under the assumptions in the question.

QUESTION 4 [1+1+2+2 = 6 marks]

Consider an American put option on a non-dividend paying stock when the stock price is $30, the strike price is $32, the risk-free rate is 5%, the volatility is 30%, and the time to maturity is 1.5 years. The option last traded at $2.50

a. What is the option's time value?
b. Explain the sign (positive or negative) of the time value you calculated in a. above. I.e., justify why the time value of this option has the sign you found above
c. Assume now that the company issuing the stock unexpectedly announces that in 1 year it will pay a dividend for the first time. How do you think the dividend announcement will affect the option premium? Explain your answer
d. Consider now a new American put option (not the one above) with zero time value and positive intrinsic value. What would a time value of zero indicate? What is the value of an option with zero time value?

SOLUTION

a) The option's intrinsic value is $32-30=2.00$. → Time value = 2.50 – 2.00 = 0.50

b) Since the put is in-the-money, its value certainly exceeds the intrinsic value → Time value is positive

c) All other things equal, the put premium is expected to increase following the announcement of the unexpected dividend. This is because the dividend payment will make the underlying asset price drop: such drop, in turn, makes the put option more valuable,

d) A time value of zero would indicate that it is optimal to exercise the option immediately. In this case the value of the option would equal its intrinsic value.

Question 5 (10 marks)

A non-dividend paying stock has a current price of $12.00. Its volatility is 30%pa and the continuously compounded risk-free rate is 4.00%pa for all maturities. A European put option on the stock has a strike of $12.50 and an expiry of six months.

a) **(4 marks)** Use a two-period binomial model to value the option today. Do all calculations to four or more decimal places. Show all your working.

First need to compute the value of the binomial rates of return over a three-month period for a stock with a volatility of 30%pa:

$u = e^{\sigma\sqrt{3/12}} = 1.1618$ (student may keep more than 4 decimal places here and their final answer may differ slightly)

$$d = \frac{1}{u} = 0.8607.$$

The risk-neutral probability of an up move (per quarter) is:

$p = \frac{e^{0.01} - 0.8607}{1.1618 - 0.8607} = 0.4960$ (student may keep more than 4 decimal places here and their final answer may differ slightly)

Next, the value of the stock at each node is:

		16.1974
	13.9416	
12		12.0000
	10.3284	
		8.8897

and the option value at expiry is:

$$f_{uu} = 0, f_{ud} = 0.5000, f_{dd} = 3.6103$$

The option value is:

$$f_0 = e^{-2\times0.04\times0.25} \times (p^2.0 + 2p(1-p)0.50000 + (1-p)^2 3.6103)$$
$$f_0 = 1.1440$$

b) **(3 marks)** Suppose you are an option market maker and have just taken a long position in one such six-month European put option. What position in the underlying stock should you take today to hedge your exposure to movements in the stock price between today and the end of the first period in the binomial tree?

Need to compute the option values at the two intermediate nodes of the tree:

$$f_u = e^{-0.01} \times (0.4960 \times 0 + (1 - 0.4960) \times 0.5) = 0.2495$$
$$f_d = e^{-0.01} \times (0.4960 \times 0.5 + (1 - 0.4960) \times 3.6103) = 2.0470$$

$$\Delta_0 = \frac{f_u - f_d}{uS - dS} = -0.4975$$

The market maker needs to take a long position of 0.4975 stocks for every long put option in their inventory.

c) **(3 marks)** Now consider the same option market maker at the end of the first period of the tree (t=1). Assuming the stock price follows the binomial distribution, does the Δ from part b) hedge their exposure from t=1 to t=2? Explain your answer.

The delta at contract initiation will only hedge the exposure from t=1 to t=2 if the ratio of the numerator to the denominator for computing Δ as expressed above is unchanged in all subsequent nodes of the tree. These ratios depend on the realization of the asset price from period 0 to period 1. If the first move is up, then the delta is:

$$\Delta_u = \frac{0 - 0.5}{16.1974 - 12} = -0.1191$$

and if it is down, this is:

$$\Delta_d = \frac{0.5 - 3.6103}{12 - 8.8897} = -1.00$$

neither of which are -0.4975. The market maker needs to adjust their hedge over time to have no exposure to the stock price.

Question 6 (10 marks)

Answer all parts of the question below.

i. A non-dividend paying stock has an expected return of 16% p.a. and a volatility of 35% p.a. The current price is $38.

a) What is the probability that a European call option on the stock with an exercise price of $40 and a maturity date in six months will be exercised? Assume the stock price follows a log normal distribution. Keep four decimal places for your final answer. Show all your working. **(5 marks)**

b) What is the probability that a European put option on the stock with the same exercise price and maturity will be exercised? Keep four decimal places for your final answer. Show all your working. **(1 mark)**

ii. What is the main difference in the underlying assumptions of the Binomial option pricing model and the Black-Scholes option pricing model? **(1 mark)**

iii. Under what conditions do the option prices from the Binomial model and the Black-Scholes model converge? **(1 mark)**

iv. What is the link between the Binomial model and the Black-Scholes model? You do not need to show any formulae in answering this question. **(2 marks)**

Solution:

a) $\ln S_T \sim \phi\left[\ln 38 + \left(0.16 - \frac{0.35^2}{2}\right)0.5, \left(0.35\sqrt{0.5}\right)^2\right]$

$\ln S_T \sim \phi[3.686961,\ \ 0.06125]$

Since $\ln 40 = 3.689$, the required probability is

$$1 - N\left(\frac{3.689 - 3.687}{0.247}\right) = 1 - N(0.01)$$

From normal distribution tables N(0.01) = 0.5040 so that the required probability is 0.4960.

b) In this case the required probability is the probability of the stock price being less than $40 in six months time. It is

1-0.4960=0.5040

ii. What is the main difference in the underlying assumptions of the Binomial option pricing model and

the Black-Scholes option pricing model?

Solution: Binomial model assumes discrete time periods/stock price movement while Black-Scholes model assumes continuous stock price movement.

iii. Under what conditions do the option prices from the binomial model and the Black-Scholes model converge? (1 mark)

Solution: The Binomial Model and the Black-Scholes Model will converge to the same option price as the number of time steps in the binomial model increases toward infinity.

iv. What is the link between the Binomial model and the Black-Scholes model? You do not need to show any formulae in answering this question. (2 marks)

Solution: Both models use risk-neutral pricing. Under both models, options are priced by taking the expected value of future payoffs, discounted at the risk-free rate.

Question 7 (10 marks)

Answer all parts of the question below.

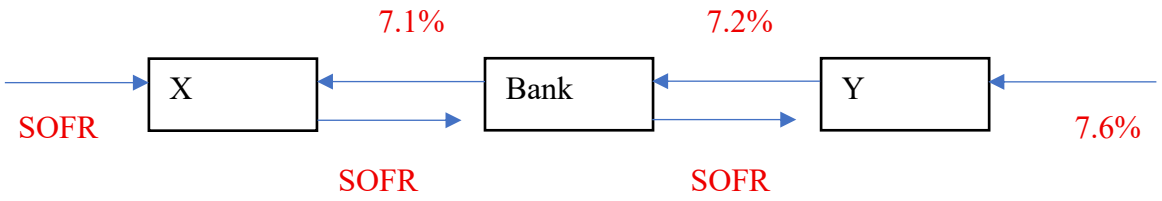
i. Companies X and Y have been offered the following rates p.a. on a $10 million 5-year investment:

	Fixed Rate	Floating Rate
Company X	7.0%	SOFR
Company Y	7.6%	SOFR+0.1%

Company X requires a fixed-rate investment; company Y requires a floating-rate investment. Design a swap that will net a bank, acting as intermediary, 0.1% p.a. and X shares 25% of the remaining gains and Y shares 75% of the remaining gains. Draw a swap diagram and show all your working. **(6 marks)**

Solution:

The spread between the interest rates offered to X and Y is 0.6% p.a. on fixed-rate investments and 0.1% p.a. on floating-rate investments. This means that the total benefit from the swap to all parties is 0.5%. Of this, 0.1% p.a. goes to the bank, 0.1% to Company X, and 0.3% to Company Y. As a result, Company X should receive a fixed-rate return of 7.1% p.a., while Company Y should receive a floating-rate return of SOFR + 0.4% p.a. The required swap is illustrated below.



ii. A 10-year, 8% coupon bond is currently trading at $90. A 10-year, 4% coupon bond is currently trading at $80. Both bonds have a face value of $100 and pay coupons semi-annually. What is the 10-year zero rate with continuous compounding? Keep four decimal places for your final answer. Show all your working. (Hint: Consider taking a long position in two of the 4% coupon bonds and a short position in one of the 8% coupon bonds.) **(4 marks)**

Solution:

Taking a long position in two of the 4% coupon bonds and a short position in one of the 8% coupon bonds leads to the following cash flows

$$\text{Year}0 : 90 - 2 \times 80 = -70$$

$$\text{Year}10 : 200 - 100 = 100$$

because the coupons cancel out. $100 in 10 years time is equivalent to $70 today. The 10-year rate, R, (continuously compounded) is therefore given by

$$100 = 70e^{10R}$$

The rate is

$$\frac{1}{10}\ln\frac{100}{70} = 0.0357$$

or 3.57% per annum.

Question 8 (10 marks)

Consider an investor with a portfolio of options. The option positions are written against the same stock (Elon Ltd). The investor seeks to delta hedge her exposure dynamically over the next two weeks. Once initiated, the hedged position will be held to the end week 1. At this point it will be re-balanced and the revised hedge position held until the end of week 2. At this point the hedge will be closed out.

The option positions (written against Elon Ltd) are

Position 1: European call, Strike $20, maturity 5 weeks, Short 20 options

Position 2: European put, Strike $23, maturity 3 weeks, Long 30 options

Elon Ltd stock is currently trading at $22. Its expected volatility is 35% p.a. The risk-free rate is 4% p.a. Assume the BSM assumptions hold.

a) How many units of the stock should the investor buy or sell to delta hedge the exposure on initiation of the hedge? Calculate the number of shares in whole numbers. **(5 marks)**

b) Now assume the stock price increases to $24 at the end of week 1. What re-balancing (if any) should occur? Calculate the number of shares in whole numbers. **(5 marks)**

Solution:

a)

Position 1:	S	22	d1	0.9679
	K	20	N(d1)	0.8340
	T	0.096154		
	n	20		
	vol	0.35		
	r	0.04		
Position 2	K	23	d1	-0.4593
	T	0.057692	N(d1)-1	-0.6772
	n	30		

Delta of portfolio =20*-0.834+30*-0.6772 =-36.9960

Therefore achieve delta neutrality via long 36.9960≈37 units of stock

b)

Position				
1:	S	24	d1	1.958433
	K	20	N(d1)	0.9750
	T	0.076923		
	n	20		
	vol	0.35		
	r	0.04		

Position				
2	K	23	d1	0.676769
	T	0.038462	N(d1)-1	-0.2483
	n	30		

Delta of portfolio =20*-0.975+30*-0.2483 =-26.9490

Therefore need to short 10 units of stock.

The new stock position will be 37-10 = 27 units of stock